Text Mining Final Assignment Report

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1 Introduction

News articles can often show bias in single sentences. Authors deliberately choose words and expressions which may contain biased attitudes and information. Such bias can influence readers' opinion and affect discussions and reduce trust towards journalism. Therefore, detecting biased content is important to understand the truth in news. However, detecting bias is not easy because it often depends on implicit signals, not clear words or topics. To detect the implicit bias in a sentence, methods need to focus on both the words and the context around them.

As one of the most influential bidirectional Transformer models, BERT gained its popularity in its ability to understand language and context. Although BERT shows strong performance in NLP related tasks, we would like to examine how much it improves over simpler methods when detecting sentence-level media bias. Thus, this lead to our research rquestion - To what extent do modern deep learning architectures outperform traditional frequency-based approaches in terms of predictive performance, error patterns, and interpretability?

To answer this question, we implemented a traditional frequency-based approach: TF-IDF + Logistic Regression baseline and a modern deep learning architecture Fine-tuned BERT model using the same dataset, article-level splits, and evaluation metrics. Beyond standard performance metrics, we conduct diagnostic analyses including confusion matrices, systematic error patterns, and model interpretability. In addition, we perform a data refinement study to examine how each model responds to changes in data quality, thereby enabling a more comprehensive analysis of the trade-offs between deep learning and frequency-based approaches for sentence-level media bias detection.

The remainder of this paper is organized as follows. In **Background**, we review relevant prior work on sentence-level bias detection. The **Data** section describes the dataset used in our experiments and presents key statistics, including sample size and label distribution. The **Methods** section details our experimental approaches, pipelines and model implementation details. The **Results** section presents various data metrics and related comparisons between these two models. The **Discussion** section shows our findings, insights, model limitations, and implications. The **Conclusion** section summarizes how our study addresses the research question and what we have learnt from experiments for two models. Finally, the **Reflection** section briefly discusses how we divided our work and the usage of AI assistants in our report.

2 Background

Earlier studies on media bias have mostly focused on entire articles or news outlets. For example, Groseclose and Milyo (Tim Groseclose, 2005) measured media bias by examining how word usage aligns with political actors, while Gentzkow and Shapiro (Gentzkow and Shapiro, 2010) analyzed overall lexical patterns across U.S. newspapers. These approaches offered valuable insights but tend to capture only large-scale bias, often overlooking more subtle, sentence-level cues that can influence readers. We argue that understanding bias at this finer granularity is crucial, as it is often at the sentence level that bias subtly shapes perception.

To address this, Recasens et al.(Recasens et al., 2013) proposed linguistic models to detect biased language, focusing on features such as framing, word choice, and syntax. Similarly, Hube and Fetahu(Hube and Fetahu, 2019) examined biased framing in news, showing that bias often arises from subjective assumptions rather than sentiment alone. These studies highlight the complexity of linguistic bias, but their reliance on handcrafted features can be limiting and somewhat rigid.

More recently, neural networks have been applied to tasks closely related to sentence-level bias detection. For instance, Da San Martino et al.(Da San Martino et al., 2019; Martino et al., 2020) demonstrated that Transformer-based models can outperform traditional methods in sentence-level propaganda detection. Building on this, Lim et al.(Lim et al., 2020) created a benchmark dataset of 966 sentences from 46 news articles, annotated on a 4-point Likert scale. Their findings emphasize how subjective bias labeling can be, and how annotators' political leanings influence their judgments—underscoring the difficulty of defining "bias" objectively.

However, Lim et al.(Lim et al., 2020) explored only basic models and did not investigate Transformer-based architectures. Moreover, the performance of such models on small datasets remains uncertain. Ezen-Can(Ezen-Can, 2020) found that Fine-tuned BERT may even underperform simpler models like LSTMs when data is limited. Henning et al.(Henning et al., 2023) observed that Transformers are sensitive to class imbalance, and Raza et al.(Raza et al., 2024) highlighted the challenge of detecting subtle bias without sufficient data. Motivated by these observations, we extend Lim et al.'s setup by applying a Fine-tuned BERT and conducting a controlled comparison to assess whether deep contextual representations can improve sentence-level bias detection under small-data constraints.

3 Data

3.1 Raw Dataset and Statistics

Sora LREC 2020 Biased Sentences Dataset (Lim et al., 2020) is used and processed from raw article-level data into a sentence-level format to enable fine-grained analysis. The result are 3,925 sentences from 46 articles. During raw data processing, labels are binarized by categorizing ratings of 1 as *unbiased* (0) and ratings 2–4 as *biased* (1). To maximize the training signal, each valid annotation is treated as a distinct sample.

To prevent data leakage and ensure generalization, the dataset was split into training (80% of the total sentences) and validation (20% of the total sentences) sets at the article level. This ensures the model is evaluated on entirely unseen sources rather than different sentences from the same article. As shown in Table 1, the dataset displays a notable class imbalance, with a biased-to-unbiased ratio of approximately 2.8:1 (2,223 vs. 800 samples in the training split). This distribution is summarized in Table 1 and suggests that class-weighting strategies need to be used during model training.

Split	Articles	Sentences	Unbiased	Biased
Train	36	3,023	800	2,223
Valid	10	902	258	644
Total	46	3,925	1,053	2,872

Table 1: Initial sentence-level dataset distribution across training and validation splits.

3.2 Experimental Observations and Data Refinement

After the initial experiments, we observed that the Fine-tuned BERT's performance was similar to that of the TF-IDF + Logistic Regression baseline, falling short of our expectations. By examining the raw data ,we noticed differences in human annotations for certain sentences, where annotators assigned conflicting labels to the same text. Such disagreement introduces label noise, which may obscure subtle bias signals and hinder effective model learning.

To mitigate this issue and investigate the influence of data quality on the two models, we decide to apply a data refinement step based on annotation consensus.To ensure a fair comparison with the baseline, the refined dataset maintains the same 80/20 article-level split as the initial experiments. Specifically, we filter the dataset using the following criteria:

- **Sample Integrity**: A sentence is kept only if this sentence has at least one valid human annotation. This will exclude unlabeled or unreliable samples.

- **Consensus Requirement**: We want to ensure that more than half of the annotators support the assigned binary label;thus we apply a majority agreement threshold of 0.5 to process the raw data. This removes highly ambiguous or noisy sentences with substantial disagreement.

As a result, the label consistency is improved though this refinement significantly reduces the dataset size. The refined dataset shows a more severe class imbalance, with roughly 1:4.8 ratio in the training split, as minority-class (unbiased)

samples are more likely to be removed by the filter criteria. Nevertheless, this trade-off ensures higher-quality annotations for meaningful model comparison. The refined sentence-level data statistics are summarized in Table 2.

Split	Articles	Sentences	Unbiased	Biased
Train	36	643	111	532
Valid	10	196	38	158
Total	46	839	149	690

Table 2: Sentence-level dataset distribution after data refinement across training and validation splits.

3.3 Data Challenges and Learnings

Stage	AVG Pairwise Cohen's Kappa
Initial	0.05
Refinement	0.19

Table 3: Improvement in inter-annotator agreement.

Dataset	Unbiased	Biased	Total	Ratio
Initial	800	2,223	3,023	1 : 2.78
Refined	111	532	643	1 : 4.79

Table 4: Class distribution and bias ratio comparison during training.

Before experimenting, we calculated the Average Pairwise Cohen's Kappa across all annotators to establish a baseline for initial agreement, and subsequently for the refined subset by backtracking to original rater-level labels. Although the Kappa score improved from 0.05 to 0.19 following our refinement process3, it remains within the "slight agreement" range. The lower Kappa scores, together with the severe class imbalance (reaching a 1:4.8 ratio in the refined set),as shown in Table 4, highlights the two challenges of this task:

Firstly, media bias is highly subjective, leading to significant label noise. Even within a filtered 'gold standard' dataset, we observed differences in human annotations, where different raters often assigned conflicting labels to the same sentence. This high level of subjectivity suggests that identifying subtle bias is an extremely difficult task even for human experts, creating a natural performance ceiling for computational models.

Secondly, the models are limited by the lack of unbiased samples and the imbalanced distribution of the data. In both the original and refined datasets, 'unbiased' examples are much fewer than 'biased' ones. This scarcity makes it harder for models to really learn what neutral language looks like.

These two issues together help explain why the two models struggle. In this setting, comparing TF-IDF with Logistic Regression,which mainly picks up clear lexical cues,and Fine-tuned BERT, which can capture deeper contextual patterns, isn't just about getting higher scores. It's more about trying to pick up subtle signs of bias that are easily hidden in noisy and tricky language.

4 Methods

4.1 TF-IDF + Logistic Regression

We used a TF-IDF + Logistic Regression model as our baseline to capture surface-level cues of biased language. In this approach, each sentence is represented as a sparse vector of term importance scores, which makes the classification process fairly straightforward and interpretable, although it doesn't explicitly account for context. To prepare the data, we converted sentences into TF-IDF vectors using unigrams and bigrams, removed common English stopwords, and discarded terms that appeared in fewer than five sentences. This helps reduce noise and focus on words or short phrases that are more likely to signal biased framing, while ignoring rare or uninformative tokens.

Then, a Logistic Regression classifier is trained on these representations using balanced class weights to address the label imbalance in the dataset as show in in Table1. By inversely weighting classes according to their frequency in the training set, the model is discouraged from over predicting the majority class and is thus encouraged to maintain sufficient sensitivity toward unbiased sentences. The maximum number of iterations is set to 1000 to ensure convergence.

To further examine model behavior, misclassified samples are analyzed alongside their prediction confidence scores, enabling qualitative inspection of systematic error patterns. This analysis provides insight into the types of linguistic constructions that remain challenging for contextual models and highlights both the strengths and limitations of Transformer-based approaches for bias detection.

Following pipeline summarizes the key steps in TF-IDF + Logistic Regression.

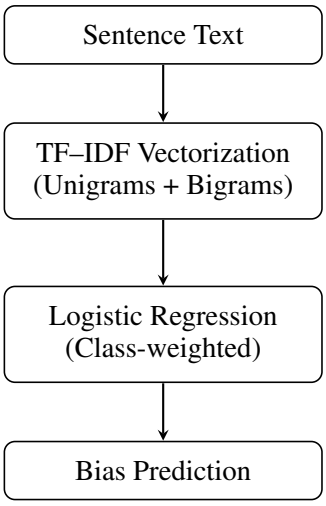


Figure 1: TF-IDF + Logistic Regression pipeline for sentence-level bias detection.

4.2 Fine-tuned BERT Model

To extract the contextual as well as the semantic cues rather than just the surface level information, we chose to use a Fine-tuned BERT model. Specifically, the binary classification for sentence-level bias classification using the `bert-base-uncased` model is used. The input for the classification task is converted into its embedding form using its particular WordPiece tokenizer. Truncation/padding has been applied to ensure all inputs are limited to 128 tokens. The attention masks are utilized to separate the padded elements from their actual counterparts.

The model is fine tuned from end to end using the same article-level training and validation splits as the TF-IDF + Logistic Regression to ensure a fair comparison. Given the observed class imbalance in the dataset, a class-weighted cross-entropy loss is used during training, assigning higher penalty to misclassification of the minority class. Class weights are computed from the training data using inverse class frequency. This encourages the model to avoid baising itself towards predicting only one class and improves sensitivity to unbiased sentences.

Training is performed for eight epochs using the AdamW optimizer with a learning rate of 2×10^{-5} and a batch size of 16. Weight decay is applied to reduce overfitting.While there is no early stopping,after training the best model checkpoint is selected based on validation F1-score. Model training and evaluation are conducted using the HuggingFace `Trainer` framework. No manual feature engineering is used,so the model learns task-relevant representations directly from raw text.

Evaluation is done by reporting the same things as the baseline model, accuracy, precision, recall, and F1-score at the sentence level. In addition to aggregate metrics, class-wise performance is analyzed to assess how well the model distinguishes between biased and unbiased sentences. To better understand model behavior, the misclassified samples are printed along with their prediction confidence scores, enabling qualitative inspection of systematic error patterns. This provides insight into the types of linguistic constructions that remain challenging for contextual models and highlights the strengths and limitations of transformer based approaches for bias .

Following pipeline summarizes the key steps in Fine-tuned BERT with Logistic Regression.

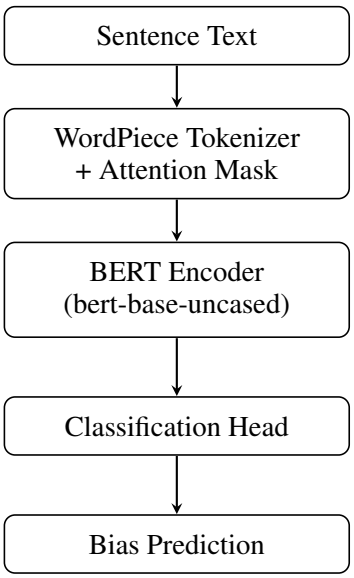


Figure 2: Fine-tuned BERT architecture for sentence-level bias detection.

4.3 Implementation Details

- **Data Partitioning:** The dataset was split into training and validation sets using an 80/20 ratio at the article level. This ensures that sentences from the same article do not appear in both sets, preventing potential data leakage.
- **Hardware:** The TF-IDF + Logistic Regression experiment was conducted on a MacBook Pro (M1, 16 GB RAM) with Python 3.11. The BERT fine tuning was conducted on Google Colab,this allowed for the usage for Google's TPUs for efficient training*(v5e-1 TPU)*.
- **Reproducibility:** To ensure reproducibility,

the random seed has been fixed at 42 for all experiments.

5 Results

5.1 Performance Metrics

Table 5 shows the classification performance of the TF-IDF + Logistic Regression model. The model achieves an overall accuracy of 0.580. For the "Unbiased" class, precision, recall, and F1-score are 0.329, 0.450, and 0.380, respectively, while for the "Biased" class, they are 0.741, 0.632, and 0.682. Macro-averaged and weighted-averaged scores are also reported.

Class	Precision	Recall	F1-score	Support
Unbiased	0.329	0.450	0.380	258
Biased	0.741	0.632	0.682	644
Accuracy		0.580		902
Macro avg	0.535	0.541	0.531	902
Weighted avg	0.623	0.580	0.596	902

Table 5: Classification performance report for TF-IDF + Logistic Regression.

Table 6 summarizes the classification performance of the Fine-tuned BERT model. The model achieves an overall accuracy of 0.645. For the "Unbiased" class, precision, recall, and F1-score are 0.377, 0.368, and 0.373, respectively, while for the "Biased" class, they are 0.749, 0.756, and 0.753. Macro-averaged and weighted-averaged scores are also reported.

Class	Precision	Recall	F1-score	Support
Unbiased (0)	0.377	0.368	0.373	258
Biased (1)	0.749	0.756	0.753	644
Accuracy		0.645		**902**
Macro Avg	0.563	0.562	0.563	902
Weighted Avg	0.643	0.645	0.644	902

Table 6: Classification performance report for the Fine-tuned BERT model on raw data.

5.2 Confusion Matrix

Figure 3 shows the confusion matrix for the TF-IDF + Logistic Regression model, illustrating the number of correctly and incorrectly classified sentences for each class. Figure 4 shows the confusion matrix for the Fine-tuned BERT, presenting similar information.Fine-tuned BERT model shows a better performance in identifying biased content, achieving a higher true positive rate of 0.76 compared to 0.63 for the TF-IDF + Logistic Regression model.

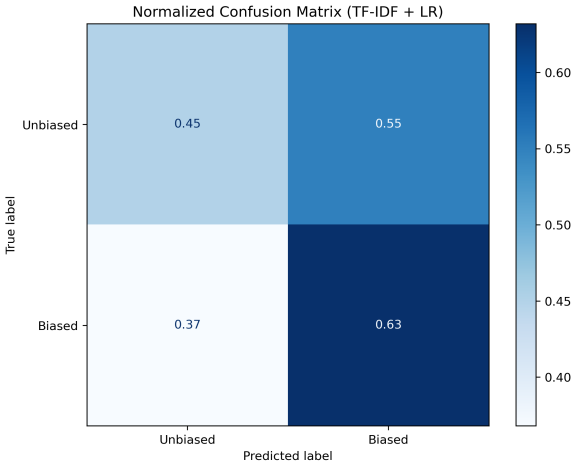


Figure 3: TF-IDF + Logistic Regression Confusion Matrix.

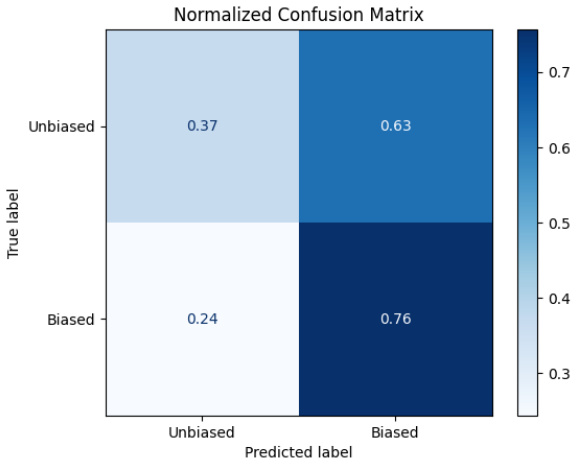


Figure 4: Fine-tuned BERT Confusion Matrix.

5.3 Error Analysis

Table 7 and Table 8 list the top-5 most confident misclassified sentences by the TF-IDF + Logistic Regression model, showing the true label, predicted label, and model confidence for each sentence.Full sentences are provided in the Appendix.

The TF-IDF + Logistic Regression model shows higher error confidence (up to 0.79) across both classes. On the other hand, the fine-tuned BERT model's top errors are primarily false positives with lower confidence scores(up to 0.55).

Sentence (truncated)	True	Predict	Confidence
We're ready to ...	1	0	0.79
Also, the ...	0	1	0.78
Pick up ...	0	1	0.77
Others, like ...	1	0	0.75
he added ...	1	0	0.75

Table 7: Top-5 most confident misclassified sentences in TF-IDF + Logistic Regression. Full sentences are in Appendix A.1.

Sentence (truncated)	True	Predict	Confidence
Miranda Richm...	0	1	0.55
Paul Ham ...	0	1	0.54
The allerga ...	0	1	0.53
The story ...	0	1	0.51
Back when ...	0	1	0.51

Table 8: Top-5 most confident misclassified sentences in Fine-tuned BERT. Full sentences are in Appendix A.1.

5.4 Interpretability

Table 9 and Table 10 show the top 10 words that each model considers most important for detecting biased sentences.

For TF-IDF + Logistic Regression, the weights are static and global. Each word gets a fixed importance score based on how strongly it correlates with the biased class across the dataset. In Table 9, the top three words are mostly specific entities or topical keywords, like "facebook" (1.266), "political" (1.200), and "trump" (1.138).

Fine-tuned BERT, on the other hand, assigns dynamic, context-dependent weights derived from attention. A word's influence on the final prediction changes depending on the surrounding words in the sentence. In Table 10, the top words are more functional or action-oriented, such as "tweeted" (0.637), "program" (0.300), and "think" (0.229).

Rank	Word	Weight
1	facebook	1.266
2	political	1.200
3	trump	1.138
4	data	1.123
5	example	1.075
6	cambridge	0.948
7	death	0.948
8	19	0.911
9	12	0.887
10	analytica	0.828

Table 9: Top 10 lexical indicators for biased sentences identified by the TF-IDF + Logistic Regression model.

Rank	Word	Weight
1	tweeted	0.637
2	program	0.300
3	think	0.229
4	hacks	0.210
5	players	0.200
6	nuclear	0.184
7	watchers	0.178
8	tillerson	0.176
9	north	0.173
10	incident	0.156

Table 10: Top 10 lexical indicators for biased sentences identified by the Fine-tuned BERT model.

5.5 Data Refinement Study

Tables 11 and 12 compare model performance on raw vs. refined data. For TF-IDF + Logistic Regression, overall accuracy increases slightly from 0.580 to 0.587, with gains mainly in the "Biased" class. For Fine-tuned BERT, overall accuracy improves from 0.645 to 0.786, with the largest gains observed in the "Biased" class as reflected in precision, recall, and F1-score.

Metric	Raw	Refined	Gain (Δ)
Unbiased (0)			
Precision	0.329	0.228	-0.101
Recall	0.450	0.474	+0.024
F1-score	0.380	0.308	-0.072
Biased (1)			
Precision	0.741	0.829	+0.088
Recall	0.632	0.614	-0.018
F1-score	0.682	0.705	+0.023
Overall Acc	0.580	0.587	+0.007

Table 11: TF-IDF + Logistic Regression Performance: Raw vs. Refined Data.

Metric	Raw	Refined	Gain (Δ)
Unbiased (0)			
Precision	0.377	0.423	+0.046
Recall	0.368	0.289	-0.079
F1-score	0.373	0.344	-0.029
Biased (1)			
Precision	0.749	0.841	+0.092
Recall	0.756	0.905	+0.149
F1-score	0.753	0.872	+0.119
Overall Acc	0.645	0.786	+0.141

Table 12: Fine-tuned BERT Performance: Raw vs. Refined Data.

6 Discussion

6.1 Performance Metrics and Confusion Matrix Analysis

The classification results in Tables 5 and 6 show that the Fine-tuned BERT model performs better overall than the TF-IDF + Logistic Regression baseline, with an accuracy of 0.645 compared to 0.580. But this improvement isn't the same across both classes, and looking closer at the metrics and confusion matrices reveals an interesting trade-off.

1. Impact of class imbalance The validation set is heavily imbalanced (28.6% Unbiased vs. 71.4% Biased), which makes both models lean toward the majority class. Even though BERT is more powerful, it actually has lower recall for the Unbiased class (0.368) than the TF-IDF + Logistic Regression baseline (0.450). From the confusion matrices, we can see that Fine-tuned BERT misclassifies 163 out of 258 unbiased sentences (63.1

2. Sensitivity and misclassification Fine-tuned BERT seems more "aggressive" in spotting bias, achieving a higher true positive rate (0.76) than the baseline (0.63). But this comes at a cost: it produces more false positives, labeling 163 unbiased sentences as biased, compared to 142 for the TF-IDF model.

3. Trade-off interpretation So while Fine-tuned BERT improves overall accuracy, its Macro-F1 score (0.563) is only slightly better than the baseline. This suggests that most of the accuracy gain comes from doing better on the majority class, rather than a balanced improvement across both biased and unbiased sentences.

6.2 Error Analysis

Looking at the errors in Table 7 and Table 8, we can see that the two models struggle in different ways. TF-IDF + Logistic Regression often misses bias when the sentence sounds neutral, since it relies mostly on keywords and doesn't really grasp the full meaning. Fine-tuned BERT, on the other hand, can be a bit too sensitive. It sometimes labels neutral, factual sentences as biased, especially when they mention investigative groups or specific organizations. This suggests that BERT might be picking up on patterns that happen to appear in biased training examples, rather than actually understanding whether a sentence is biased.

Overall, these observations show the trade-offs between simpler keyword-based models and deeper contextual ones. TF-IDF + Logistic Regression is safer but can miss subtle bias, while BERT catches more bias but occasionally over-interprets neutral content. This makes it clear that, even with strong models, detecting bias at the sentence level is still tricky—especially when subtle context and nuanced language come into play.

6.3 Interpretability Analysis

Tables 9 and 10 show what kinds of features each model relies on to detect bias. The TF-IDF + Logistic Regression model mostly picks up topic-related keywords. For example, words like "facebook" (1.266), "political" (1.200), and "cambridge" (0.948) are tied to specific events. Even numbers like "19" and "12" get relatively high weights. This suggests that the model is focusing more on the subject of a sentence rather than on how bias is actually expressed.

Fine-tuned BERT, in contrast, seems to pay more attention to action words or functional language, like "tweeted" (0.637) and "think" (0.229). Some political names, like "tillerson," (0.176) still appear, but with lower weights. This suggests that BERT is more sensitive to how statements are framed,

instead of just spotting controversial entities. In practice, this could help it generalize better across different news topics, rather than getting tied down to specific events.

6.4 Data Refinement Study Analysis

We compared the models on both the raw dataset and a refined version that only included sentences where most annotators agreed (threshold ≥ 0.5).

For TF-IDF + Logistic Regression (Table 11), the refinement didn't make much difference. Accuracy went up slightly from 0.580 to 0.587, mostly due to better performance on the biased class (F1 +0.023, precision +0.088). However, the unbiased class actually got worse, with F1 dropping 0.072 and precision down 0.101. This suggests that the baseline model doesn't really gain much from cleaner labels and still mostly depends on keywords instead of understanding subtle sentence meaning.

Fine-tuned BERT (Table 12) showed a very different pattern. It clearly benefited from the refined dataset: overall accuracy jumped from 0.645 to 0.786. The biased class improved a lot in recall (+0.149) and F1 (+0.119), while the unbiased class had higher precision (+0.046) but a small drop in recall (-0.079). What's interesting is that this happened even though the training set got much smaller : from 3,023 to 643 sentences, a 78.7% reduction. This shows that BERT can really take advantage of higher-quality labels to pick up subtle signals of bias.

Overall, these results suggest that while TF-IDF barely benefits from refining annotations, Fine-tuned BERT is much more sensitive to label quality and can generalize better when trained on more consistent data.

7 Conclusion

In this study, we investigated to what extent modern deep learning architectures, specifically Fine-tuned BERT, outperform traditional frequency-based methods like TF-IDF + Logistic Regression in terms of predictive performance, error patterns, and interpretability. Our experiments show that Fine-tuned BERT consistently performs better at capturing context and subtle bias cues at the sentence level. On the raw dataset, BERT already outperformed the baseline, and when we refined the dataset to include only sentences with high annotator agreement, its performance improved even more, despite the much smaller training size.

The error analysis highlights the trade-offs between the two approaches. TF-IDF + Logistic Regression is easy to interpret but often misses subtle bias, while BERT catches more nuanced signals but can sometimes over-interpret neutral content. This shows that the advantage of deep learning models depends on both the quality of the data and the task requirements.

Overall, our study suggests that modern deep learning models are powerful for sentence-level bias detection, but traditional methods still have value when interpretability is important or labels are noisy. For future work, it would be interesting to explore ways to reduce BERT's false positives, perhaps by using multi-task learning or integrating external knowledge—so that the model can better distinguish truly neutral reporting from biased statements.

8 Reflection

The work was divided such that Jian Song implemented the TF-IDF + Logistic Regression baseline model, while Rahul Ray implemented the Fine-tuned BERT model. The report was written jointly through iterative discussion which led us to conduct another set of experiments with data refinement. No AI tools used in the writing process itself. LLM assistance(Claude sonnet 4.5) was used only for formatting TikZ diagrams (Rahul) and was checked for correctness with the actual text in the diagrams input by Rahul Ray.

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A Example Appendix

A.1 Complete List of Sentences

Top-5 most confident misclassified sentences in TF-IDF + Logistic Regression.

- "We're ready to talk any time they'd like to talk," he said.(Confidence: 0.79)
- Also, the insult "bum" has been dragged back into the lexicon, which is nice, given that it will probably irk Trump as much as any other (accurate) label. (Confidence:0.78)
- Pick up and leave. (Confidence:0.77)
- Others, like Republican Reps. Wesley Morgan and Jeff Hoover, seemed to echo Ham's sentiment. (Confidence:0.75)
- he added. (Confidence:0.75)

Top-5 most confident misclassified sentences in Fine-tuned BERT based model.

- Maranda Richmond, who did not respond to TIME's request for comment on Thursday, told the news organization that Johnson molested her on New Year's Eve in 2012 when he was a pastor at the Heart of Fire Church. (Confidence:0.55)
- Paul Ham, the chairman of the Bullit County GOP, criticized the Kentucky Center for Investigative Reporting in a Thursday interview with the Washington Post. (Confidence:0.54)
- The allegations against Johnson surfaced in an investigation published by the Kentucky Center for Investigative Reporting. (Confidence:0.53)
- "The story was based on hearsay: no arrest, no conviction, no jail time," he said.(Confidence:0.51
- "Back when the Constitution meant something, a man could stand before a jury of his peers. (Confidence:0.51)